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Disulfiram enhances meropenem activity against NDM- and IMP-producing carbapenem-resistant *Acinetobacter baumannii* infections

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Objectives: To evaluate the *in vitro* and *in vivo* efficacy of the FDA-approved drug disulfiram in combination with meropenem against MBL-expressing carbapenem-resistant *Acinetobacter baumannii*.

Methods: Chequerboard and antibiotic resistance reversal analysis were performed using 25 clinical isolates producing different MBLs. Three representative strains harbouring NDM, IMP or non-MBL genes were subjected to a time–kill assay to further evaluate this synergistic interaction. Dose-dependent inhibition by disulfiram was assessed to determine IC_{50} for NDM-1, IMP-7, VIM-2 and KPC-2. Further, to test the efficacy of meropenem monotherapy and meropenem in combination with disulfiram against NDM- and IMP-harbouring *A. baumannii*, an experimental model of systemic infection and pneumonia was developed using BALB/c female mice.

Results: Chequerboard and antibiotic reversal assay displayed a synergistic interaction against MBL-expressing *A. baumannii* strains with 4- to 32-fold reduction in MICs of meropenem. In time–kill analysis, meropenem and disulfiram exhibited synergy against NDM- and IMP-producing carbapenem-resistant *A. baumannii* (CRAb) isolates. *In vitro* dose-dependent inhibition analysis showed that disulfiram inhibits NDM-1 and IMP-7 with IC_{50} values of 1.5 ± 0.6 and 16.25 ± 1.6 μM, respectively, with slight or no inhibition of VIM-2 (<20%) and KPC-2. The combination performed better in the clearance of bacterial load from the liver and spleen of mice infected with IMP-expressing CRAb. In the pneumonia model, the combination significantly decreased the bacterial burden of NDM producers compared with monotherapy.

Conclusions: These results strongly suggest that the combination of disulfiram and meropenem represents an effective treatment option for NDM- and IMP-associated CRAb infections.

Introduction

Acinetobacter baumannii has emerged as an important global pathogen, owing to its high propensity for severe epidemics and multidrug resistance.[1] Consequently, the WHO has listed this pathogen as a high-priority pathogen for the development of new therapeutics.[2] Carbapenems are considered as one of the 'last resort' drugs that exhibit a broad range of activity against untreatable infections with Gram-negative pathogens.[3] Regrettably, the perpetual efficacy of carbapenems is being threatened by the global rise of determinants of β-lactam resistance. Amongst the most worrisome determinants are the MBLs, which are capable of inactivating all classes of β-lactams. In addition to non-susceptibility to carbapenems, MBL-expressing strains tend to harbour other determinants of resistance, the most common being resistance to aminoglycosides and fluoroquinolones.[4,5] Therefore, treatment options for patients suffering from infection by MBL-positive pathogens are extremely limited, leading to the increased use of colistin, which exhibits high toxicity.[6]

Antibiotic monotherapy is a conventional strategy to combat bacterial infections, but combination therapy is not exceptional.[7,8] The combination of antibiotics with an adjunct has been very successful in clinics, exemplified by Augmentin and Avycaz. These combinations are composed of a β-lactam antibiotic (amoxicillin and ceftazidime, respectively) with a β-lactamase inhibitor (clavulanate and avibactam, respectively).[9,10] There are numerous inhibitors against serine β-lactamases, like sulbactam, tazobactam and clavulanate, which have subsequently improved the activity spectrum of β-lactam antibiotics. MBLs are a mechanistically distinct class of β-lactamases that harbour zinc atom(s) at their active site to aid the hydrolysis of β-lactam rings. So far, no clinically approved inhibitor of MBLs has been discovered.[11] Discovery of inhibitors of MBLs has become a serious unmet need in clinics, which has gained significant breadth in the identification of compounds to target these enzymes. Over the years, several reports about metal-chelator adjunct molecules that display *in vitro* activity against MBLs have been published. However, due to high host-associated toxicity, they have failed to demonstrate the same *in vivo*.[12]

Disulfiram (Antabuse) is a drug used primarily as a deterrent to aid patients in developing abstinence in cases of chronic alcoholism.[13] In one study, it was shown that disulfiram forms dithiocarbamate metal complexes with zinc/copper and acts as a proteasome complex inhibitor. It is currently in clinical trials for cancer treatment.[14] Further, disulfiram's ability as an *in vitro* enzyme inhibitor for NDM has been highlighted recently.[15] In the present study, we sought to investigate disulfiram's ability as an adjunct to potentiate the antibacterial activity of meropenem against MBL-positive isolates. Disulfiram displayed inhibition of clinically relevant MBLs, notably NDM and IMP. Furthermore, we tested the meropenem/disulfiram combination in a murine systemic infection and pneumonia model of infection and showed that it can sensitize both NDM- and IMP-expressing clinical isolates of carbapenem-resistant *A. baumannii* (CRAb) to meropenem.

Materials and methods

Bacterial strains and plasmids

MBL-positive clinical strains of Gram-negative pathogens were obtained from patients attending AIIMS Rishikesh, India and other clinical isolates were obtained from Government Medical College and Hospital, Chandigarh, India as mentioned in our previous study.[7] All MBL-positive strains displayed MICs of meropenem in the range 8–128 mg/L. Disulfiram demonstrated an MIC of >256 mg/L for all isolates. The class B β-lactamase-encoding genes, including bla_{NDM}, bla_{IMP}, bla_{VIM}, bla_{GIM} and bla_{SIM} were detected using PCR with gene-specific primers (Table S1, available as Supplementary data at *JAC* Online). The standard strain of *A. baumannii* AYE was obtained from ATCC, University of Boulevard, Manassas, VA, USA. Plasmid pET28b expressing NDM-1, VIM-2, IMP-7 or KPC-2 were a kind gift from Prof. Gerry Wright, McMaster University, Hamilton, Canada. All growth experiments were performed in CAMHB from HiMedia. Unless stated otherwise, all chemicals and reagents were purchased from Sigma–Aldrich. All antibiotics were dissolved in Milli-Q water; disulfiram was dissolved in DMSO.

MIC determination

All bacterial isolates were allowed to grow overnight with shaking at 180 rpm at 37°C in 4 mL of CAMHB. These saturated bacterial cultures were subsequently diluted 1:100 into fresh CAMHB and grown to an OD_{600} of 0.4 for growth. Bacteria were again diluted 1:1000 in CAMHB containing the final concentration of the drug. The plates were incubated at 37°C under stationary conditions for 16 h and OD_{600} was measured after 16 h.

Chequerboard analysis

Bacterial isolates were allowed to grow overnight with shaking at 180 rpm at 37°C in 4 mL of CAMHB. Saturated bacterial cultures were subsequently diluted 1:100 in fresh CAMHB and grown to an OD_{600} of 0.4 for growth. Bacteria were again diluted at 1:1000 in CAMHB and added to drug assay plates. The plates were incubated at 37°C and OD_{600} was measured after 16 h. An assay for each compound combination was used to calculate the FIC index (FICI). FICI values were interpreted as follows: ≤0.5, synergy; >0.5–1.0, additive; >1.0–4.0, no interaction; and >4.0, antagonism.

Time–kill assays

Time–kill assays were performed in CAMHB using CRAb. Overnight cultures of *A. baumannii* RPTC 2, RPT C4 and RPTU 60 were diluted 1:100 in fresh CAMHB and grown at 37°C. Different concentrations of test compounds were added to the culture, adjusted to 10^4 cfu/mL, and incubated at 37°C with shaking at 180 rpm. Aliquots of culture were periodically sampled (100 μL), 10-fold serially diluted in 1× PBS and plated onto Mueller–Hinton (MH) agar plates. Plates were incubated overnight at 37°C. Time–kill curves were plotted using means of colony count ($\log_{10}$ cfu/mL) versus time over 24 h.

Dose–response curve

Carbapenemases (final concentrations: NDM-1, 50 nM; IMP-7, 10 nM; VIM-2, 20 nM and KPC-2, 5 nM) were incubated with 30 μM nitrocefin, after a 15 min pre-incubation with disulfiram (in the range 0–100 μM). The buffer used for enzyme assays was 50 mM HEPES (pH 7.5) with 0.01% Tween-20 (Chelex treated). To monitor the rate of hydrolysis of nitrocefin, samples were read in 96-well microplate format using a SpectraMax reader (Molecular Devices) at 490 nm (30°C).

In silico docking simulation

The Protein Data Bank (PDB) files of NDM-1 and VIM-2 structures were obtained from the RCSB PDB (NDM-1 PDB ID: 6TGD; VIM-2 PDB ID: 7A5Z).[16,17] The structure of IMP-7 MBL was not available on PDB and was obtained from the SWISS-MODEL repository (UniProt: Q93AU3) that had been previously modelled using the template PDB ID: 5EV8.[18] The surface area and cavity volume of each of the MBLs were predicted using CASTp, which computes Richard's solvent-accessible surface area and volume of the clefts/cavities in a protein.[19] The active-site pocket cavity was selected, and the gap region void was visualized using the SURFNET program in Chimera.[20,21] The ligand structure of disulfiram was downloaded from the ZINC drug database.[22] All the molecular structures were preprocessed using PyMOL, and AutoDock Tools was used to convert the PDB files of MBLs and ligand structure into PDBQT format.[23,24] For performing molecular docking of disulfiram with the surface of MBLs, AutoDock Vina was used with parameters of exhaustiveness and num_modes set to 40 and 10, respectively.[25] The grid box was set to cover the active-site cavity with one of the zinc atoms as centre coordinates. The protein–ligand interactions were computed by LigPlot$^+$ and were subsequently visualized by using Chimera.[26]

Ethics

The animal studies were approved by the Institutional Animal Ethics Committee (IAEC) and performed according to the submitted protocols (BT/IAEC/2017/03 and BT/IAEC/2017/21).

In vivo murine systemic infection model

Ten-week-old female BALB/c mice weighing 20–22 g were used. Neutropenia in mice was chemically induced by administrating 150 mg/kg cyclophosphamide at Day −2 and −1 of infection. To establish systemic infection, 10^9 cfu/mL *A. baumannii* RPTC 2 in 0.1 mL PBS was injected into neutropenic mice via IV injection. After 6 h post-challenge, mice were treated every 12 h with meropenem (10 mg/kg), disulfiram

Table 1. MIC and FICI for meropenem, disulfiram (DSM) and their combination against Gram-negative clinical isolates with different MBL genes

		MIC (mg/L)				
Clinical isolates	MBL genes	Meropenem	DSM	Meropenem with 14 mg/L DSM	Fold change	FICI
A. baumannii						
RPTU 59	NDM	16	>256	4	4	0.2812
RPTU 60	NDM	16	>256	4	4	0.2812
RPTU 61	NDM	8	>256	2	4	0.2812
RPTU 62	NDM	8	>256	0.5	32	0.0625
RPTC 2	IMP	32	>256	8	4	0.2812
RPTC 5	IMP	8	>256	2	4	0.2812
RPTC 6	IMP	16	>256	4	4	0.2812
RPTC 10	IMP	16	>256	2	16	0.0937
RPTC 23	IMP	16	>256	4	4	0.2812
RPTC 28	IMP	64	>256	8	8	0.1562
RPTC 35	IMP	64	>256	16	4	0.2812
RPTC 34	IMP	128	>256	32	4	0.2812
RPTC 27	IMP	64	>256	16	4	0.2812
RPTC 36	NDM	32	>256	8	4	0.2812
RPTC 32	IMP	16	>256	2	4	0.2812
RPTC 29	IMP	32	>256	8	4	0.2812
AYE	NA	1	>256	0.5	2	1.0000
P. aeruginosa						
RPTCPA 68	IMP	64	>256	16	4	0.2812
RPTCPA 69	NDM	64	>256	8	8	0.1562
RPTCPA 70	IMP	128	>256	16	8	0.1562
RPTCPA 71	NDM	32	>256	4	8	0.1562
RPTCPA 72	NDM	32	>256	8	4	0.2812
RPTCPA 73	IMP	64	>256	16	4	0.2812
RPTCPA 74	IMP	64	>256	16	4	0.2812
E. coli						
RPTCEC 1	NDM	8	>256	0.5	16	0.0937
RPTCEC 2	NDM	16	>256	2	8	0.1562

(7 mg/kg) or the meropenem/disulfiram combination (10 and 7 mg/kg, respectively). The mice in the untreated group were injected intraperitoneally with PBS. Disulfiram was dissolved in 5% DMSO, 30% polyethylene glycol and 0.01% Tween 80 in 1× PBS. Following treatment, all mice from each group were sacrificed to assess the bacterial burden in the spleen and liver.

In vivo murine pneumonia model

Neutropenic female mice were anaesthetized with an intraperitoneal injection of ketamine (100 mg/kg) and xylazine (10 mg/kg) and infected with 10^9 cfu/mL *A. baumannii* RPTU 60 in 30 μL of PBS intranasally. Post-challenge, mice were intraperitoneally injected with meropenem (10 mg/kg) or the meropenem/disulfiram combination (10 and 7 mg/kg, respectively) at 2 and 8 h. After 24 h post-infection, lungs were homogenized in PBS, serially diluted and plated onto the MH agar plates. Harvested left lungs were fixed using 10% formalin and implanted in paraffin, followed by haematoxylin and eosin staining. Tissue sections from each lung sample were viewed under a Zeiss microscope.

Statistical analysis

Statistical data analyses were performed using Prism 8 software (GraphPad, CA, USA). Student's *t*-test was used for comparison between two groups. Time–kill assay was analysed using one-way ANOVA (Kruskal–Wallis) to compare combination therapy with the sum of the effects of each agent alone. P values of <0.05 were considered statistically significant.

Results

Among 105 different Gram-negative clinical isolates, 25 isolates exhibited intrinsic carbapenem resistance due to MBLs, based on the difference in the zones of inhibition (ZOIs) of meropenem in the presence and absence of EDTA (exhibiting a difference in ZOI of >6 mm) and were selected for further experiments. A total of 10 isolates were NDM producers and 15 were IMP producers, as verified by different MBL-specific primers. No VIM-, GIM- or SIM-producing isolates were found.

Chequerboard and MIC reversal analysis

The MICs of meropenem and disulfiram and FICI of meropenem in combination with a fixed concentration of disulfiram (14 mg/L) are presented in Table 1. All isolates demonstrated high MICs of disulfiram (>256 mg/L). Synergy was found for 25 out of 25 (100%) clinical isolates expressing NDM and IMP. Among the 25 isolates showing synergistic interaction, disulfiram at a fixed concentration of 14 mg/L reduced the MIC of meropenem by 4- to 32-fold for NDM-positive isolates and 4- to 16-fold for IMP-expressing isolates. Upon treatment with disulfiram at 14 mg/L, most isolates (18/25, 72%) were susceptible to meropenem (MIC $\leq$8 mg/L). Non-carbapenem-resistant *A. baumannii* AYE displayed additive interaction with an FICI of 1. We also checked whether potentiation by disulfiram is specific to the carbapenem class of antibiotics or it can display synergy with other antibiotics as well. Disulfiram potentiated the activity of doripenem, another carbapenem class antibiotic (Figure S1 and S2), and conversely, no potentiation was observed between non-carbapenem β-lactam, quinolone, monobactam or aminoglycoside antibiotics and disulfiram.

Time–kill assays

Time–kill assays revealed that the combination of meropenem and disulfiram was extremely bactericidal and exhibited better killing against NDM- and IMP-harbouring *A. baumannii* than meropenem alone at the same concentration. For 4 mg/L meropenem in combination with 16 mg/L disulfiram, synergy was obtained at 4 h for NDM-positive RPTU 60, which was maintained for 24 h. Treatment with meropenem alone at 4 mg/L did not display significant killing until the timepoint of 24 h (Figure 1a). A similar pattern of killing was observed for IMP-harbouring *A. baumannii* RPTC 2, suggesting that disulfiram is effective in eradicating both NDM- and IMP-positive *A. baumannii* isolates (Figure 1b). In contrast, the combination displayed ineffective killing of non-MBL CRAb RPTC 4 (Figure 1c). Further, we assessed the potential for emergence of resistance in NDM- and IMP-harbouring clinical isolates of *A. baumannii* at 2× MIC of meropenem. We observed the values of frequency of resistance in *A. baumannii* RPTU 60 and RPTC 2 to be 8.9×10^{-6} and 2.8×10^{-6}, respectively. In combination with disulfiram, meropenem did not yield any resistant mutants, thus suggesting a lower propensity for resistance selection.

In vitro inhibition of MBLs

To confirm the effect of disulfiram on NDM-1 and IMP-7 activity, dose–response assays were conducted, taking nitrocefin as a substrate. In the absence of supplemental zinc, disulfiram inhibited NDM-1 and IMP-7 with IC_{50} values of 1.5 ± 0.6 and 16.25 ± 1.6 μM, respectively (Figure 2a and b). VIM-2 demonstrated a minor dose-dependent inhibition (i.e. <20%) (Figure 2c), whereas the serine-β-lactamase KPC-2 was not inhibited by disulfiram at the concentrations tested (Figure 2d). Collectively, the above results are indicative of preferential inhibition of NDM-1 and IMP-7 by disulfiram.

In silico docking simulation

To investigate the reason behind observed differences in IC_{50} values amongst different MBLs, surface analysis on NDM-1, IMP-7 and VIM-2 followed by *in silico* molecular docking between disulfiram and MBLs were performed. It was observed that disulfiram was interacting with the active-site clefts in all the three MBLs. The visualization of solvent-accessible cavities showed that NDM-1 has the largest active-site cavity with a volume of 777.4 Å^3. This cavity was suited to accommodate sulphur group moieties of disulfiram close to the active-site zinc ions (Figure 3a). The docking study revealed that one sulphur atom of disulfiram coordinated with two zinc ions (at distances of 2.4 and 3.5 Å) and one sulphur atom of the disulphide bond interacted with one zinc ion (at 3.5 Å), to provide a basis of dual-functional inhibition of disulfiram in NDM-1 (Figure 3b). The volume of the active-site cavity of IMP-7 at 284.1 Å^3 facilitates easy accommodation of disulfiram, which displays three coordination interactions (at 3.6, 2.6 and 3.6 Å) between the zinc ions and sulphur atoms of the inhibitor (Figure 3c and d). For VIM-2, the active-site cavity volume at 154.4 Å^3 was too narrow to facilitate proximity of the inhibitor's sulphur atoms with active site zinc ions (Figure 3e and f). Disulfiram at the cleft of VIM-2 encountered hydrophobic interactions with Phe62, Trp87, Glu146, Asp117, His179, Asn210 and Asp 118 (Figure S3). This loose interaction might result in the weak inhibitory activity of disulfiram against VIM-2.

In vivo murine systemic model

A preliminary single-dosing assay determined that disulfiram alone at 7 mg/kg and meropenem up to 50 mg/kg in spleen tissue have no effect in lowering the bacterial burden (Figure S4). Mice were infected by IV administration of IMP-harbouring *A. baumannii* RPTC 2. After treatment, bacterial load in the liver and spleen was examined. We found that treatment with meropenem had little effect on bacterial load; however, co-administration of meropenem and disulfiram resulted in a significant reduction of the bacterial burden in the liver (*****P* < 0.0001) as well as in the spleen (***P* < 0.01) (Figure 4).

In vivo murine pneumonia model

The *in vivo* efficacy of the disulfiram/meropenem combination was assessed in a mouse pneumonia model infected with NDM-harbouring *A. baumannii* RPTU 60. The bacterial burden of the lungs was evaluated after 24 h of infection after treatment. The results displayed that the bacterial load decreased in the combination-treated group by 3 log-fold (****P* < 0.001) in comparison with the group treated with meropenem alone (Figure 5a). The therapeutic effects were confirmed by histological staining. We observed pathological changes in the untreated group, including alveolar fusion, a large amount of inflammatory cell infiltration and congestion. By contrast, the disulfiram/meropenem combination treatment resulted in reduced pathological damage in the lungs (Figure 5b).

Discussion

A. baumannii, in addition to being persistent in antibiotic treatment, also harbours various determinants of resistance against an array of clinically relevant antibiotics, MBL being the most concerning.[12,27,28] MBL-harbouring *A. baumannii* strains have become a scourge in clinical settings, as their resilience to available treatment strategies has allowed them the opportunity to thrive.[29] In this work we have evaluated the potential of an FDA-approved dithiocarbamate derivative, disulfiram, to improve the antibacterial efficacy of meropenem against NDM- and IMP-producing Gram-negative pathogens. Chequerboard assays demonstrated that synergistic interaction exists between disulfiram and meropenem or doripenem against all clinical isolates

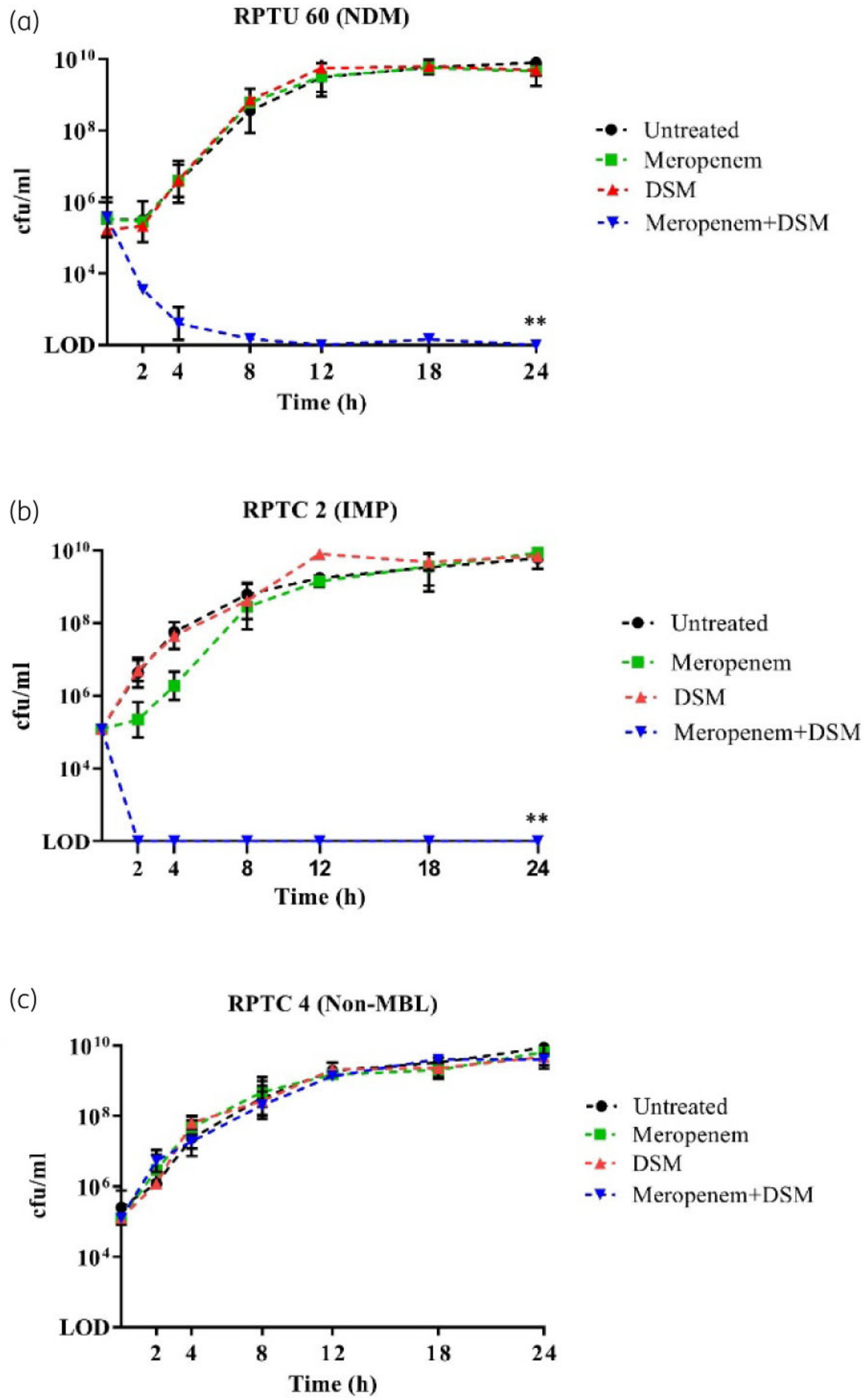

Figure 1. Disulfiram (DSM) inhibits *in vitro* activity of NDM and IMP. The potent synergy of meropenem and disulfiram is demonstrated by time–kill curves, which show that the population of (a) NDM- and (b) IMP-harbouring strains at the exponential phase is significantly lowered (more than 1000-fold) upon exposure to the combination of meropenem and disulfiram. Disulfiram enhanced the potency of meropenem towards MBL-harbouring strains (a and b) but not against non-MBL-harbouring strain (c). Mean values of three independent replicates are shown and error bars indicate the SD. Limit of detection (LOD) was assigned to values of <200 cfu/mL. ** signifies $P<0.01$ for the combination group compared with meropenem or DSM treatment alone at the end of 24 h. This figure appears in colour in the online version of *JAC* and in black and white in the print version of *JAC*.

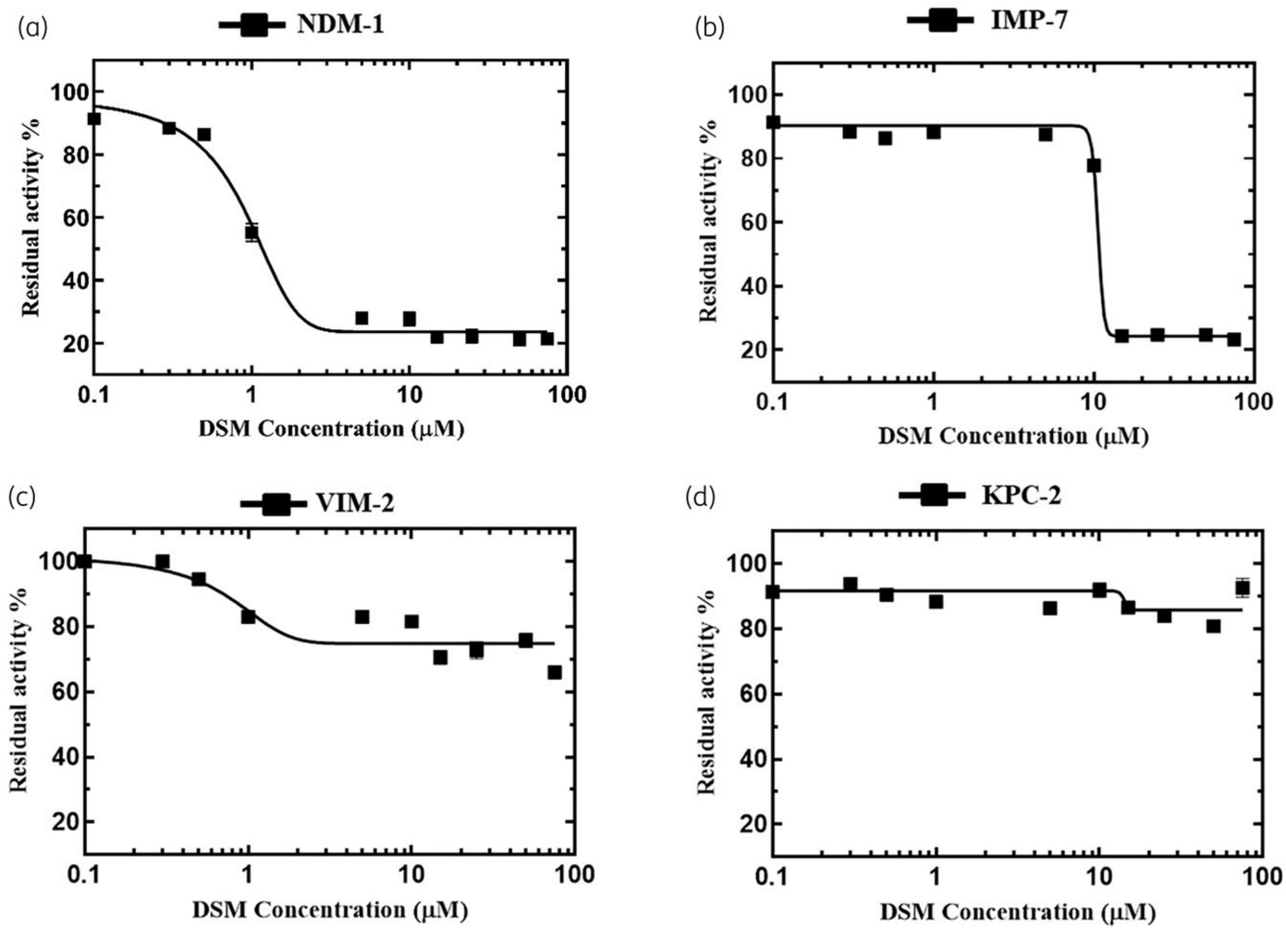

Figure 2. Disulfiram (DSM) shows dose-dependent inhibition of purified enzymes NDM-1 and IMP-7. NDM-1 and IMP-7 showed dose-dependent inhibition with observed IC_{50} values of 1.5 ± 0.6 and 16.25 ± 1.6 μM, respectively (a and b). VIM-2 showed minimal inhibition (c), whereas non-MBL KPC-2 was unaffected by disulfiram (d). Error bars denote the SD of at least three technical repeats.

of *A. baumannii*, *Pseudomonas aeruginosa* and *Escherichia coli* expressing NDM and IMP determinants of MBL. In the time–kill assays, the combination displayed synergy in isolates expressing NDM and IMP up to 24 h, but not against non-MBL-expressing CRAb isolates. A dose-dependent inhibition assay to determine the IC_{50} of disulfiram against NDM-1, IMP-7, VIM-2 and KPC-2 showed that disulfiram is potent in inhibiting nitrocefin hydrolysis activity of NDM-1 and IMP-7. This finding is consistent with a previous study, which suggested disulfiram as a promising NDM-1 inhibitor by various protein- and mutation-based assays and highlighted its dual-functional inhibitory potential against MBLs.[15] Nevertheless, disulfiram's ability to inhibit MBLs and to potentiate meropenem in *in vivo* infection models has not been evaluated. In this study, murine systemic and pneumonia models of infection also confirmed the *in vivo* potential of this combination. Our results demonstrated that disulfiram is a promising adjunct that resensitizes NDM- and IMP-positive *A. baumannii* to the action of meropenem.

MBL-harbouring *A. baumannii* infection is one of the most dreaded and hard-to-treat infections afflicting the medical community worldwide. Rather than continuing with the small molecule-based drug discovery approach, an alternate strategy to identify a compound that serves as an adjunct to available antibiotics is the unmet need of the hour.[30] MBLs harbour Zn(II) at their active site, which facilitates the hydrolysis of the carbapenem class of antibiotics. However, the usage of successful metal-targeting chelators results in grave toxicity concerns, as proven in unsuccessful therapeutic attempts using EDTA.[12]

Chelators are known to be non-specific molecules that are capable of forming a metal complex with a large number of ions.[31] Such promiscuity in metal binding is among the most prominent feature of chelators limiting their role in healthcare settings.[32] Disulfiram acts on NDM-1 by chelation of its active site Zn(II), and can also covalently bind by forming a disulphide bond with its Cys208 residue.[15] Disulfiram is approved by the FDA for administration up to 500 mg daily.[33] The maximum serum level of disulfiram when 250 mg is administered orally is 8.094 μM, and upon its hepatic metabolism the produced metabolite, diethyldithiocarbamate (DTC), is also a divalent metal ion chelator and a potent inhibitor of MBLs with IC_{50} values of 3.9 ± 1.6 and 12.5 ± 3.8 nM for NDM-1 and IMP-1, respectively.[15,34]

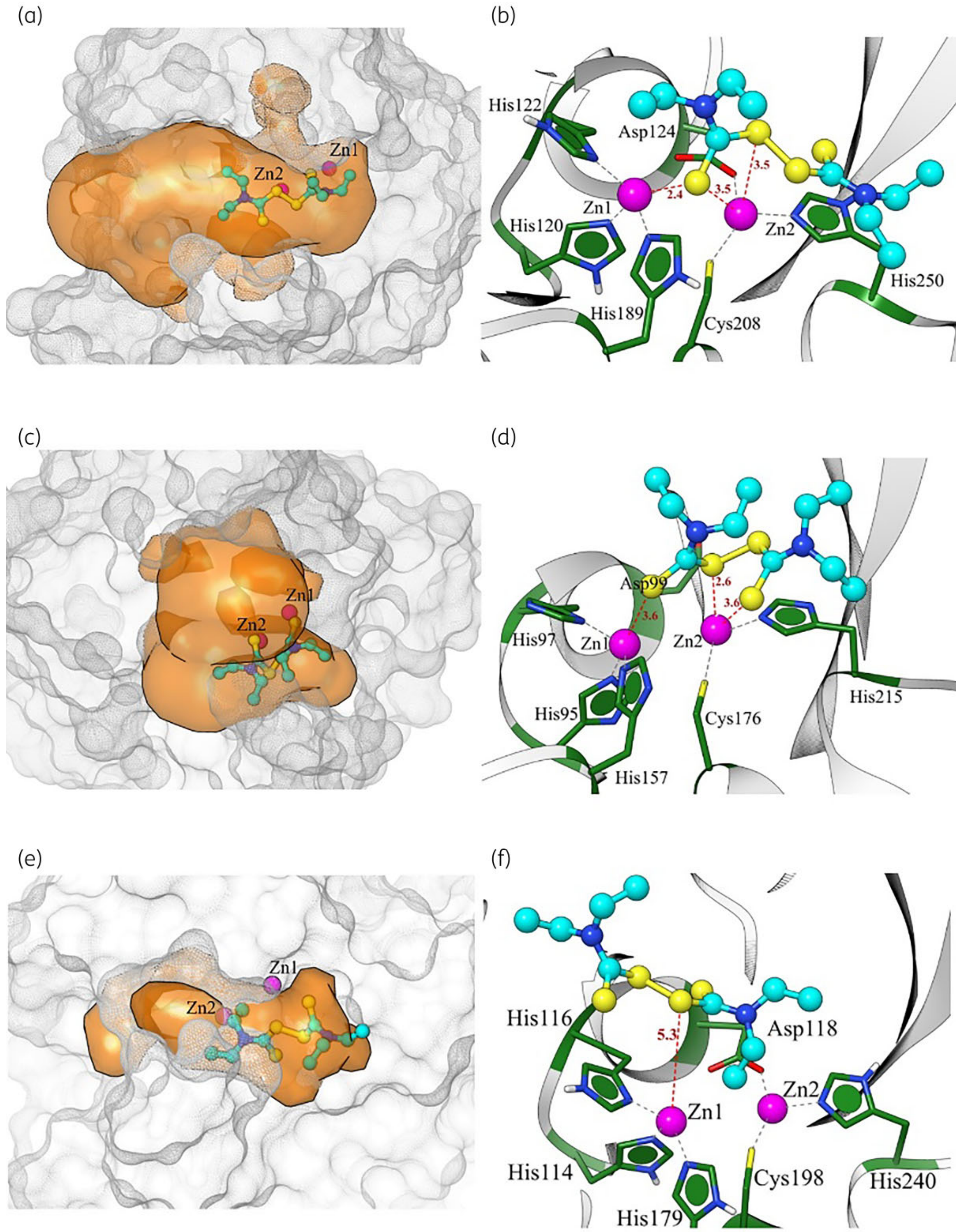

Figure 3. Active-site cavity volume and *in silico* docking of MBLs with disulfiram. NDM-1 (a) has the largest active-site pocket volume of 777.4 Å^3, facilitating efficient interaction (b) between sulphur group moieties of disulfiram and active-site zinc ions. The cavity volume of IMP-7 (c) is 284.1 Å^3, resulting in three coordination bond interactions (d) between disulfiram and zinc ions. The narrow active-site cavity of VIM-2 (e), with only 154.4 Å^3 volume, resulted in poor/weak interaction (f) of disulfiram with the active-site zinc ions. Each enzyme has two Zn ions in the active-site region, shown as magenta spheres. The carbon, nitrogen and sulphur atoms in disulfiram are shown in cyan, blue and yellow, respectively. The interactions between disulfiram and MBLs are represented as red dashed lines. This figure appears in colour in the online version of *JAC* and in black and white in the print version of *JAC*.

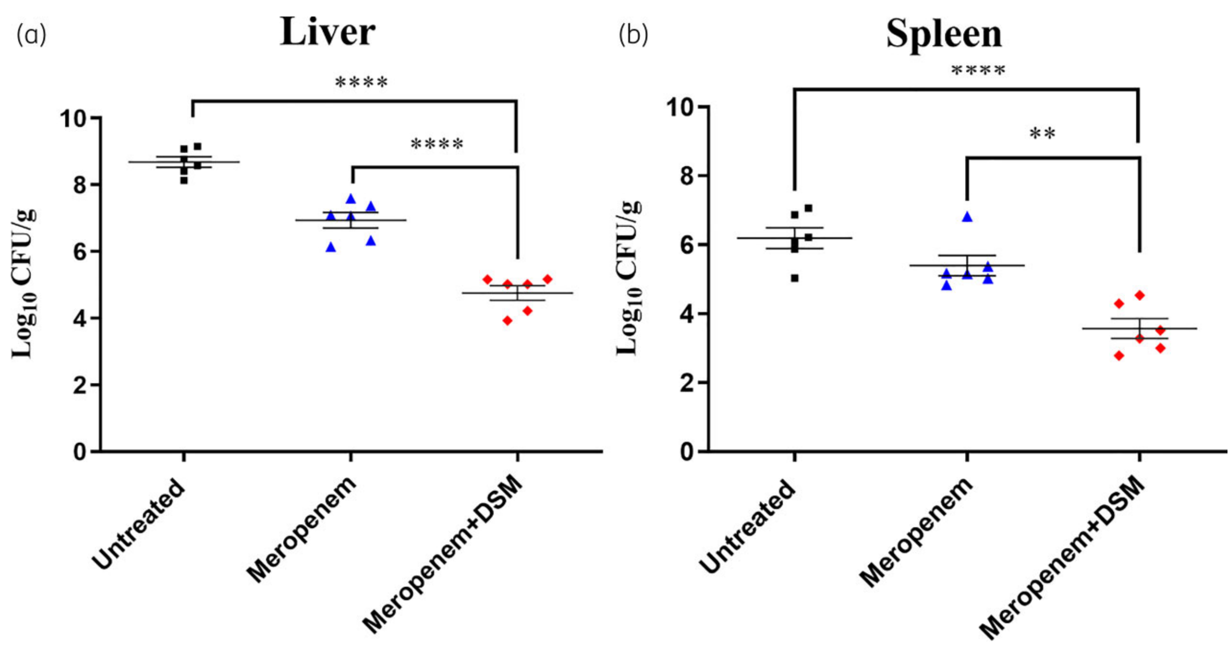

Figure 4. *In vivo* antibacterial activity of meropenem (10 mg/kg) and the combination of 10 mg/kg meropenem with 7 mg/kg disulfiram (DSM) in a systemic infection model against IMP-1-positive clinical isolate *A. baumannii* RPTC2. Bacterial load was measured in the liver (a) and spleen (b) after 48 h. The data are presented as mean ± SD, based on values obtained from six mice. Differences were considered statistically significant from the untreated group when $P < 0.05$. The P value was calculated using Student's t-test. ****$P < 0.0001$, **$P < 0.01$. This figure appears in colour in the online version of *JAC* and in black and white in the print version of *JAC*.

Disulfiram is usually taken orally in 1–3 divided doses of 100–500 mg daily, and the maintenance dose is 100–200 mg.[35,36] Acute toxicity can occur in patients with doses higher than 500 mg/dL.[37] Previous studies on pharmacokinetic parameters have shown that it has a half-life of 7.3 h.[34] The toxicity of disulfiram has been broadly investigated previously, which demonstrated no indication for mutagenic, teratogenic or carcinogenic effects.[37,38] No toxicity has been documented with disulfiram at its suggested dosage and no other serious adverse effects have been associated with disulfiram.[39] Although disulfiram was initially introduced in healthcare settings as an alcohol deterrent, recent important findings have highlighted its antibacterial properties against many Gram-positive pathogens. Disulfiram as an antibacterial agent not only inhibits the growth of *Staphylococcus aureus* and *Francisella tularensis* (MIC: 4–16 mg/L) but also displays synergy with vancomycin and improves the latter's bactericidal activity against vancomycin-resistant *S. aureus* (VRSA).[40,41] Sheppard *et al.*[42] demonstrated that the *S*-octylthio derivative of disulfiram possesses effective growth-inhibitory activity against *Staphylococcus*, *Streptococcus*, *Bacillus*, *Enterococcus* and *Listeria* spp. One study proposed that disulfiram can be repurposed to target infections caused by non-tuberculous *Mycobacterium* infections, as it shows synergy with all antibiotics except linezolid and meropenem against *Mycobacterium fortuitum* and *Mycobacterium abscessus*.[43] These studies indeed positioned disulfiram as a candidate drug to revive the existing scenario of antibiotic treatment failure due to Gram-positive pathogens. However, disulfiram's potency against Gram-negative pathogens has not been fully explored. Our results demonstrate that disulfiram in combination with meropenem inhibits NDM-1 and IMP-7 at micromolar concentrations against all clinical isolates expressing NDM and IMP, and a substantial decrease in the MIC of meropenem for carbapenem-resistant pathogens. The activity of the meropenem was restored to a susceptible range (i.e. <8 mg/L) in around 72% of all clinical isolates tested.

Although different MBLs hydrolyse carbapenems in the same manner, their affinity for disulfiram is varied, which results in inhibition of NDM-1 and IMP-7, but not VIM-2. This is also evident from the difference in IC_{50} values of disulfiram against IMP-1 (2.8 ± 0.8 μM)[15] and IMP-7 (16.25 ± 1.6 μM). *In silico* docking analysis demonstrated that due to the large active-site cavity volume of NDM-1 and IMP-7, disulfiram can easily interact with their active-site zinc ions and thereby can inhibit the hydrolysis of meropenem and nitrocefin. The relatively weak/loose interaction between disulfiram and VIM-2 due to the narrow active-site cavity contributes to the overall poor interaction of sulphur groups of the inhibitor and zinc ions of VIM-2. This finding corresponds well with the results obtained in the dose-dependent inhibition assays. Additionally, in our systemic infection and pneumonia murine model, 10 mg/kg meropenem was ineffective in lowering the bacterial burden in the liver, spleen and lungs. Administration of meropenem with 7 mg/kg disulfiram greatly improved the former's efficacy, thereby lowering the bacterial titre ≥2 log-fold with respect to the group treated with meropenem alone. These experiments were performed three times and showed consistent results.

In conclusion, disulfiram can restore the *in vitro* and *in vivo* efficacy of meropenem against NDM- and IMP-expressing *A. baumannii*. This study proposes disulfiram as a promising adjunct contender in the treatment of NDM- and IMP-expressing *A. baumannii* infections.

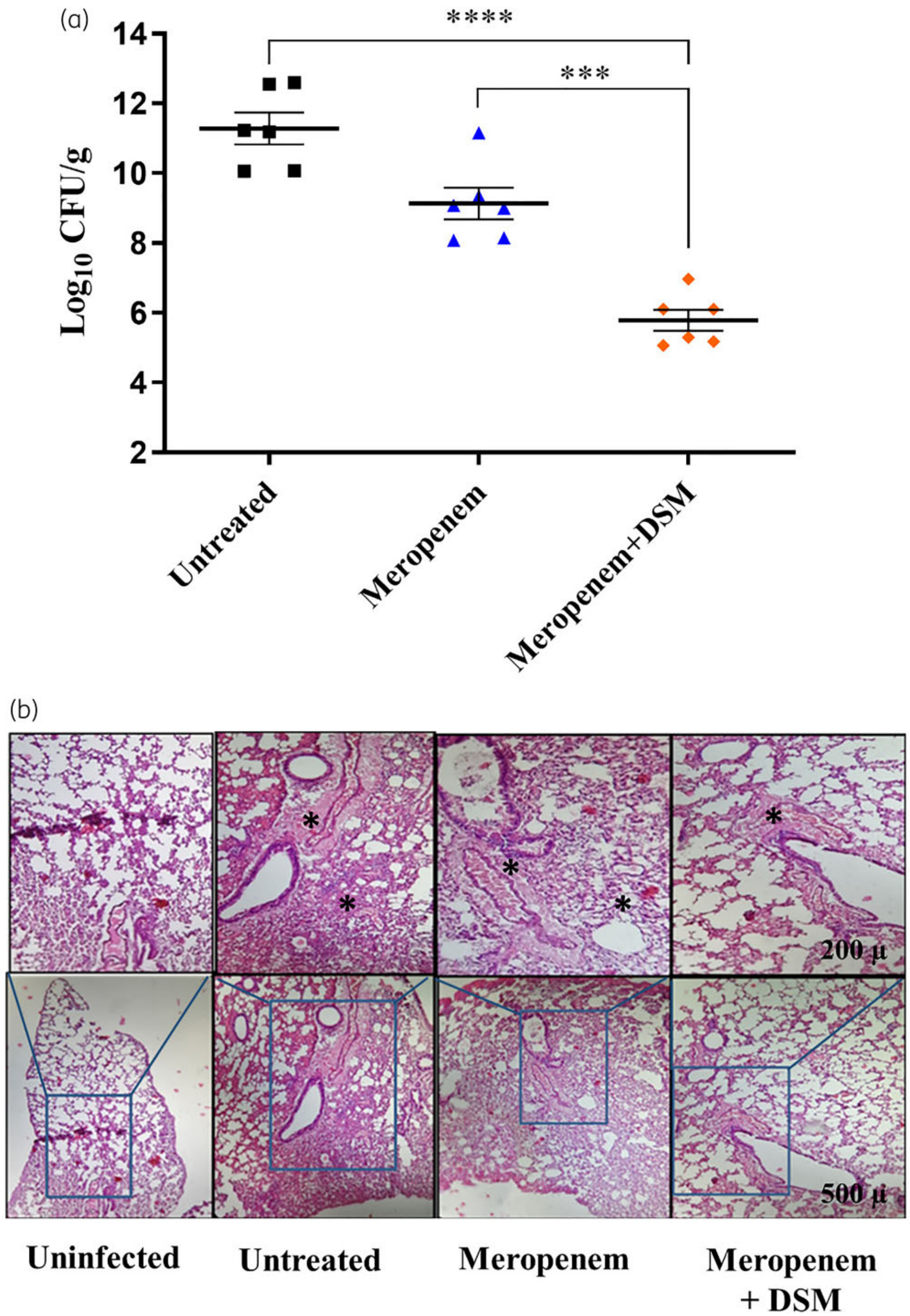

Figure 5. (a) *In vivo* antibacterial activity of meropenem (10 mg/kg) and the combination of 10 mg/kg meropenem with 7 mg/kg disulfiram (DSM) in a mice pneumonia infection model against NDM-1-positive clinical isolate *A. baumannii* RPTU60. The data are presented as mean ± SD, based on values obtained from six mice. Differences were considered statistically significant from the untreated group with $P < 0.05$. The *P* value was calculated using Student's *t*-test. ***$P < 0.001$; ****$P < 0.0001$ (b) Lung histology was done using haematoxylin and eosin staining in the mice pneumonia model after treatments. * denotes regions of alveolar pus aggregation. This figure appears in colour in the online version of *JAC* and in black and white in the print version of *JAC*.

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Transparency declarations

None to declare.

Author contributions

Conceptualization: V.D.; data curation: V.D.; formal analysis: V.D., G.K., M.S. and R.P.; funding acquisition: R.P.; investigation: R.P.; methodology: V.D., K.D., V.K.G. and G.K.; project administration: R.P.; resources: R.P., A.K. and B.J.O.; software: V.D. and G.K.; supervision: R.P.; validation: V.D. and R.P.; writing (original draft): V.D.; writing (review and editing): V.D., M.S. and R.P. All authors approved the final version of the manuscript.

Supplementary data

Materials and methods for *in vitro* frequency of resistance and enzyme purification, Table S1 and Figures S1 to S4 are available as Supplementary data at *JAC* Online.

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